COGS 9: Introduction to Data Science
Group Project Checkpoint #2 (10 pts) - Due Tuesday, Mar 17 @ 11:59pm | *NO FINAL EXAM*
Submit through Gradescope as a group, DO NOT submit individually!

Group Member Information

First Name	Last Name	PID
Wei	Yen	
Mark	Bishoff	
Nuraiym	Mirbekova	
Ricardo	Campos Murillo	
Jessica	Lu	

Data Science Question

- Do artists who are active between 2018 to 2025 whose music has higher average danceability achieve higher average popularity on Spotify?

Hypothesis

- Artists who were active between 2018 and 2025 and produced songs with higher average danceability will, on average, achieve higher levels of popularity on Spotify.
- This hypothesis is based on the intuition that highly danceable music tends to perform better on streaming platforms like Spotify as they are more upbeat and easy to listen to, making them more likely to be added to playlists, played at social events, and replayed multiple times. Spotify's algorithm through their auto-generated playlists and recommendations often promote songs that generate stronger listener engagement which can be more easily achieved by more danceable songs. In addition, short-form video platforms such as TikTok and Instagram Reels have become more popular during that time frame. Songs that trend/are more likely to be used on these platforms emphasize rhythm and beat, characteristics captured by more danceable music. These trends during this time period suggest a positive relationship between an artist's average danceability and their average popularity on Spotify.
-

Background Information

- Predicting which songs will become popular has long been a challenge in music analytics, but accurately predicting how songs become hits or misses is a bigger challenge than it seems. Traditional approaches that rely on musical features, lyrics, or self-reported listener

preferences have generally shown low predictive accuracy. This problem is often referred to as the "Hit Song Science" problem (McFee). In this source researchers proposed building models of neurophysiologic responses to determine hit songs from the flops, and they found emotional and there are key connections between emotional and neural activation and popularity. For example, one study demonstrated that while linear models using neural measures classified hit songs with about 69% accuracy, ensemble machine learning models capturing non-linear patterns achieved classification accuracies exceeding 95%. These findings suggest that emotional engagement with music, rather than conscious self-reports plays a key role in determining popularity, supporting the idea that measurable musical attributes associated with emotional response (such as danceability) may be meaningfully related to a song's success.

- In another research paper, researchers compared different music genres using a variety of audio features such as tempo, energy, danceability, acousticness, and valence, analyzing their relationships with song popularity through t-tests and correlation analysis (Singh). The study concluded that **different features influence popularity depending on the genre**. For example, pop songs generally tend to achieve higher popularity scores, and there is a strong positive correlation between certain musical attributes, particularly energy and loudness, with a correlation coefficient of 0.822. By examining the behavior of five audio features relative to a song's popularity across pop, rock, hip-hop, and country genres, the analysis identified both general and genre-specific patterns. These findings have important implications for music production, as they help clarify which musical factors are most strongly associated with commercial success across different genres.

Data

- Our ideal perfect data set would have both metrics to measure artist popularity and danceability of the songs. It would also include all artists from 2018 to 2025. I would need as many observations as there are artists from 2018 to 2025. I would need variables like artists name, song name, danceability score, and number of streams.
- There is a kaggle data set that I found online: https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset
- It has about 30000 unique artist values. They already collected track_id, artist name, album name, track name, danceability, popularity, duration, explicit and energy.
- The data set differs from our ideal data set in several ways. We only took from one source which is spotify, and not considering factors like soundcloud and youtube. They also didn't include the total streams, as that number is constantly fluctuating. But we do have instead a popularity metric to measure from. There are also no engagement data that could be helpful. The data set had more than what I needed, but also not perfect.
-

Ethical Considerations

- (0.5pt) Data Collection
- We need to consider collection bias and informed consent. The dataset we're using contains Spotify tracks across 125 genres with audio features calculated by specifically Spotify's algorithms. The main issue is collection bias since our dataset only includes tracks available on Spotify, excluding artists from other platforms like SoundCloud, Apple Music, and other streaming services. This creates bias toward commercially distributed music and may underrepresent independent, underground, or regional artists and not capture music tastes of all music listeners.
- Additionally, Spotify's audio feature algorithms , like danceability, may have built-in biases in how they measure different genres. For example, the algorithm might calculate danceability differently for electronic dance music compared to classical music, potentially skewing our results. To address this, we'll acknowledge in our findings that our analysis only applies to Spotify's ecosystem and cannot be generalized to the entire music industry or all artists active between 2018-2025.
- (0.5pt) Data Storage
- Since we're using a publicly available Kaggle dataset, we don't have major privacy concerns. We'll store the data on our personal computers with basic password protection and only share it among our group members for this project. Once we're done with the analysis and submit our final report, we'll delete the dataset and only keep our final results and visualizations. We'll make sure to back up our work in case we need to verify anything, but there's no need for complex security measures since it's public information about songs and artists.
- (0.5pt) Data Analysis
- During analysis, we need to consider missing perspectives and dataset bias carefully. Our analysis focuses purely on quantitative audio features and popularity scores, but we're missing crucial contextual factors that drive artist success, like marketing budgets, record label support, playlist placements, viral social media moments and trends. To address missing features, we should examine the dataset for multiple types of bias, like if danceability calculation works equally well across all 125 genres, or does it favor certain musical styles? Is the popularity metric biased toward mainstream Western music or certain languages? Does the dataset adequately represent artists from different regions and cultural backgrounds? Our visualizations and summary statistics must represent the underlying data, we shouldn’t choose results that support our hypothesis or hide findings that contradict it. We also need to ensure our analysis is fully documented and reproducible, with clear code comments and version control.
- (0.5pt) Modeling (Statistical or Machine Learning)

- We will be sure to sample from a diverse group of artists to ensure we are not being discriminatory. That means we will be choosing music of varied genres and diverse artists. Fairness will be ensured and we'll test our model for disparate error rates and unfair predictions.
- We will be critical about how we are optimizing for our defined metrics. It's important to note that Spotify's popularity scores are just measuring recent plays, so they are not a real measure of a song's quality or artistic significance. Amazing songs could score low on danceability because they're being experimental or are arranged in a way that eluded the danceability scoring algorithm, and songs could have a low popularity but still be groundbreaking tracks that push the boundaries of music forward. It's essential to communicate with those who see our findings that we are just measuring popularity as it relates to danceability, not if danceability is definitively what makes a song good or successful. This way artists wouldn't feel pressured to incorporate danceability, while sacrificing artistry, and the public won't have a skewed perception of what makes a good song.
- We'll have to address fairness, explainability, and metric selection and ensure our model doesn't discriminate against certain genres, artist demographics, or musical styles. Testing our model's performance across different subgroups to check for disparate error rates or unfair predictions, will be important as well.
- For metric selection, we need to be critical about what we're optimizing, Spotify's popularity score is based on recent play counts, which doesn't measure artistic quality, cultural impact, or long-term significance. A song could be artistically groundbreaking but score low on popularity if it's niche or ahead of its time. We need to clearly explain our model's decisions in understandable terms and communicate its limitations extensively, that our model can only identify correlations in historical data, not predict future hits or determine what makes generalized "good" music. We should explicitly state that correlation between danceability and popularity doesn't mean that making music more danceable will automatically increase an artist's success.
- (0.5pt) Project or Model Deployment
- Once we release our findings, it's possible that the data makes it seem like danceability is a metric that will make or break an artist's song. We must be clear about the fact that our findings are just revealing a pattern, and they don't mean a song is better, has elevated artistic merit, or is more deserving of popularity. We don't want to give artists the notion that they must follow some danceability formula in order to be successful on the Spotify platform. There are many variables we are not accounting for that include but are not limited to, marketing, acclaim, relevance, and prior success. While we won't be submitting our findings to the public, it's important to acknowledge these pitfalls so that we can understand the real-world implications a study like this can have.

- For deployment, we need to think about how our findings could be misused even though this is just a class project. If the music industry used analysis like ours, record labels might only sign artists who fit a certain algorithmic profile, which would hurt musical diversity. Artists might feel pressured to make their music more danceable just to game the system instead of being creative. We need to be clear that our analysis just shows patterns in past data, it's not telling artists how to make music. Correlation doesn't mean causation, and there are tons of factors beyond danceability that make artists successful, like marketing, timing, and luck. Since we're not actually deploying this publicly, we just need to acknowledge these risks in our report to show we understand the real-world implications.

Analysis Proposal (3pts)

You will propose **how** you would use data to answer your research question. You **do NOT** need to run the analysis or report final results. Instead, focus on showing a clear understanding of the concepts from class and how you would apply them.

We are grading **conceptual correctness and appropriate method of choice**, not highly technical implementation details. For example, if you plan to use machine learning, clearly state whether your task is **regression** (predicting a numeric value) or **classification** (predicting a category).

If you need help or feedback about any part of your plan, post on Piazza, go to discussion section and PLA office hours, and/or do your own additional research (e.g., Google, Wikipedia, Stack Exchange, AI tools).

⚠️ Be aware that directly copy-and-pasting the output of an AI chatbot is a serious academic violation and will be reported.

Include **all** of the following sections in your analysis proposal:

1. **Data Collection:** Where will your data come from, and how will you gather it (API, scraping, database, open dataset, etc.)?
2. **Data Wrangling:** What steps will you take to clean, combine, and structure your data so it's ready to analyze?
3. **Descriptive & Exploratory Data Analysis:** What summary statistics, relationships (e.g., correlations), and exploratory plots will you use to understand the data and check patterns?
4. **Data Visualization:** What **explanatory** visualization(s) would you create to communicate your findings, and why is that visual appropriate?

(0.5pt) **Data Collection**

We will use the publicly available Spotify tracks datasets from Kaggle, they contain features like danceability and popularity. The dataset themselves were collected through spotify's API, and we will filter the data to limit to contain only artists that were active between 2018 and 2025. We may also

supplement the data set with other data using the Spotify Web API ourselves to verify or add additional data.

(0.5pt) Data Wrangling
We will clean the dataset by removing duplicate tracks, delete missing values, and filter songs released between 2018 and 2025. We can then group the data by artist and compute each artist's average danceability and popularity across their songs.

(0.5pt) Descriptive & Exploratory Data Analysis
First, we will compute univariate analysis for each variable, danceability and popularity. allowing us to find summary statistics of those relevant columns. Through that, it will allow us to see if there are any outliers and patterns. However, to better understand our hypothesis, we will perform a bivariate analysis, calculating the correlation (r) between those two variables, finding the strength and direction of their relationship.

(0.5pt) Data Visualization
To visualize our data, we could first create box plots (summary statistics & outliers) and histograms for danceability and popularity separately, allowing us to visualize patterns and outliers. Our main visualization will be a scatter plot to see correlation between popularity and danceability (danceability on x-axis and popularity on the y-axis) as they are both quantitative variables. It directly shows us if there is a positive or negative relationship, as well as showing us if it is very weak or strong.

Next, choose two analysis methods from the list below and describe how you would apply each one to your question:

- Statistical Analysis (inference, hypothesis tests, A/B testing, etc.)
- Predictive Analysis (machine learning; classification, regression, etc.)
- Computer Vision Techniques (image data)
- Text Analysis (sentiment analysis, TF-IDF, topic modeling, etc.)
- Time Series Analysis
- Dimensionality Reduction
- Another method that you have researched on your own

For each selected method, explain:

- what you would do (inputs, outputs, key steps),
- why it's appropriate for your question, and
- how you would interpret the results.

USE THE FOLLOWING TEXT HEADERS AND DELETE INSTRUCTIONS ABOVE BEFORE SUBMITTING. Include 2-3 sentences for each answer.

(0.5pt) Analysis Method 1 - Statistical Analysis

We will use a hypothesis test to test for the correlation between danceability and popularity. So after collecting the data of danceability and popularity from a sample of artists, we will then use that as an input to calculate the Pearson's correlation coefficient and the p-value to see whether the findings are statistically significant. We will decide whether our findings are statistically significant based on p-value interpretation.

Our null hypothesis would be that there's no correlation between an artist's average danceability and their average popularity on Spotify, resulting in p-value = 0, on the other hand our alternative hypothesis is that artists with higher average danceability also tend to have higher average popularity, meaning there is a positive relationship between the two variables giving a p-value > 0. If the p-value is below 0.05, we reject the null hypothesis and conclude there is a statistically significant positive relationship between danceability and popularity. This is appropriate because we want to see if there is meaningful correlation between danceability and popularity of music, and calculating the slope of the correlation as well as if it's significant will help us determine this.

(0.5pt) Analysis Method 2 - Predictive Analysis
We will also use linear regression to model the popularity based on the danceability. The model will have danceability as an input and predicted popularity as an output. This is appropriate because it helps us predict the popularity of songs and see the relationship between the variables. We will split the data into a training set and testing set before fitting, to evaluate generalization rather than just in-sample fit. Our primary evaluation metric will be R^2, which tells us the proportion of variance in popularity explained by danceability alone. In addition, we will also report RMSE on the test set to quantify prediction error in interpretable units. We're simply going to interpret the results by how well linear regression fits on the data and by the model error to see how well danceability predicts popularity in a linear model, and this will help us see if we need to implement a non-linear model.

Discussion (2pts)

In this section, explain what your analysis results would *mean* and how confident you could be in your conclusions.

Include all of the following:

- Interpreting results: If your analysis supports (or does not support) your research question, what would you conclude? What would count as meaningful evidence?
- Limitations, confounds, and bias: Describe the main limitations of your approach (e.g., measurement error, missing data, small sample size, assumptions of the method, overfitting, correlation vs. causation). What confounding variables and/or sources of bias in your data could affect your results (e.g., sampling bias, platform/user bias)?
- How you will address these issues: Explain specific steps you would take to reduce or evaluate these problems (e.g., robustness checks, alternative models, controlling for confounds, stratifying samples, improving data collection, validation with another source).

- Societal/ethical implications: Summarize the key societal and/or ethical concerns from your Ethical Considerations section and explain how you would mitigate them (privacy, consent, fairness, harms, transparency, responsible reporting, etc.).

ADD YOUR GROUP'S TEXT BELOW AND DELETE INSTRUCTIONS ABOVE. Include 2-3 sentences for each answer.

(0.5pt) Interpreting results

If we find a strong positive correlation and a significant p-value, we would conclude that danceability is a key factor in Spotify popularity, supporting our hypothesis. Meaningful evidence would be an R value that explains a substantial portion of the variance in popularity, showing that the trend is consistent across different genres. If no correlation is found, we would conclude that other factors like marketing or genre-specific trends are more influential than rhythm alone.

(0.5pt) Limitations, confounds, and bias

A major problem is the algorithmic bias within the danceability metric, which is an automated score rather than a metric. Because Spotify's algorithm determines this score based on specific rhythmic patterns, it may unfairly penalize genres with complex or non-Western structures, leading to a skewed relationship with popularity. Furthermore, while popularity is based on objective number/metric, the danceability score remains a "black box" that reflects the biases from the developers who programmed it.

(0.5pt) How you will address these issues

We could incorporate sensitivity analysis across regions and genres, creating danceability to popularity scores across each grouping to account for those distinctions. By normalizing these scores within specific categories, we prevent the algorithm from developing a bias toward specific dominant genres that might otherwise skew the overall dataset

(0.5pt) Societal/ethical implications

A major ethical concern is that our findings might encourage artists to "game the system" by mathematically optimizing songs for danceability rather than creative expression. This could lead to a decline in musical diversity and the devaluation of music as an art form in favor of profit. To mitigate this, we will explicitly state that our results identify historical patterns, not a definitive formula for "good" or "successful" music, elaborating further that tastes may change throughout the years, hence why we focused on songs between 2018-2025.

Group Contributions

Each group member must write 1-2 sentences describing their contributions to Project Checkpoint 2.

Students who **did not contribute at all** will receive a 0 for their checkpoint grade. To support fair grading, the person submitting the checkpoint may indicate any **non-contributing group members** below.

- __Wei ting Yen__ : Handled Data collection and data wrangling.
- __Mark Bishoff__ : Handled the Discussion Section
- __Jessica Lu__ : Worked on Analysis Proposal
- __Ricardo Campos Murillo__ : Did the Analysis Methods section
- __Nuraiym Mirbekova__ : Worked on additional details of Analysis Methods and ensured all of the information matches the rubric